

Complete Solution of $Ax=b$

Let us start with an example:

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

this has to be in a column space of A to have a solution.

$$\hookrightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & | & b_1 \\ 2 & 4 & 6 & 8 & | & b_2 \\ 3 & 6 & 8 & 10 & | & b_3 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & | & b_1 \\ 0 & 0 & 2 & 4 & | & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & | & b_3 - 3b_1 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & | & b_1 \\ 0 & 0 & 2 & 4 & | & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & | & b_3 - b_2 - b_1 \end{bmatrix}$$

$$\hookrightarrow b_3 = b_1 + b_2$$

What does this signify?

For A to have a complete solution b_3 must be equal to $b_1 + b_2$.

Basically, with this process, you can find the solvability of a matrix.

Now, we will find the complete solution.

Step 02: Find $x_{\text{particular}} (x_p)$

$\rightarrow$ Set all free variables to 0

$$\begin{bmatrix} 1 & 2 & 2 & 2 & | & 1 \\ 0 & 0 & 2 & 4 & | & 3 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

free free

$$\hookrightarrow \begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 1 \\ 2x_3 + 4x_4 &= 3 \end{aligned}$$

setting, $x_2=0$ and $x_4=0$

$$\hookrightarrow \begin{aligned} x_1 + 2x_3 &= 1 \\ 2x_3 &= 3 \end{aligned}$$

$$\hookrightarrow x_3 = 3/2$$

$$x_1 = -2$$

Assuming, $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$

$$\hookrightarrow \begin{aligned} b_1 &= 1 \\ b_2 - 2b_1 &= 3 \end{aligned}$$

$$b_3 - b_2 - b_1 = 0$$

$$\text{so } x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

Step 02: Here, $x_{\text{complete}} = x_{\text{particular}} + x_n$

$x_n = \text{Nullspace of } A$

$$\hookrightarrow Ax_p = b$$

$$Ax_n = 0$$

$$A(x_p + x_n) = b + 0$$

$$\hookrightarrow Ax = b$$

So, find the nullspace.

From step 01:

$$\left. \begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 0 \\ 2x_3 + 4x_4 &= 0 \end{aligned} \right\} \text{Remember to make them zeros}$$

$$\begin{aligned} &\text{Setting } x_2 = 0, x_4 = 1 \\ &\rightarrow 2x_3 + 4 = 0 \rightarrow x_3 = -2 \quad \text{so } \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \\ &x_1 + 0 - 4 + 2 = 0 \rightarrow x_1 = 2 \end{aligned}$$

$$\begin{aligned} &\text{Setting } x_2 = 1, x_4 = 0 \\ &\rightarrow 2x_3 = 0 \rightarrow x_3 = 0 \quad \text{so } \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \\ &x_1 + 2 + 0 + 0 = 0 \rightarrow x_1 = -2 \end{aligned}$$

$$x_n = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\therefore X_{\text{complete}} = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + C \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

But there are more into this.

The relation between rank, column and row.

Think about a $m \times n$ matrix of rank ' r '.

So what's the relation between r & m and r & n ?

$$\hookrightarrow r \leq m \quad \text{and} \quad r \leq n.$$

But there are variations.

Case 1: Full column rank, $r = n$; $r < m$ (Informally called tall and thin)

Meaning, all are independent columns.

Meaning, every column has a pivot. So, n pivots.

So, the null space will have only zero vector.

$$Ax = b$$

$$\hookrightarrow x_p + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = x_p \rightarrow \boxed{\text{so only zero on one solution exists}}$$

Case 2: Full row rank, $r = m$, $m \leq n$, $r < n$ (Short and Wide)

Example: $x + y + z = 3$

$$x + 2y - z = 4$$

$$\hookrightarrow \begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\xrightarrow{\text{REF}} \begin{bmatrix} 1 & 0 & 3 & | & 2 \\ 0 & 1 & -2 & | & 1 \end{bmatrix} = \begin{bmatrix} I & F \end{bmatrix}$$

$$\hookrightarrow x_p = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad x_n = C \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

So if a matrix is full row rank matrix,
 $Ax=b$ has a solution for all right hand side b .
 Meaning, all rows have pivots.

Their column space $\mathbb{R}^m: Ax=b$ is always solvable

Also there are $n-r = n-m$ special solutions in $\mathcal{N}(A)$

$$Ax=b$$

↳ There are infinitely many solutions.

Case 1: Full rank $r=m=n$ (Square and Invertible)

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 15 \\ 25 \end{bmatrix}$

↳ $x_p = \begin{bmatrix} 1 \\ -\frac{7}{3} \\ 1 \end{bmatrix}, x_n = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

↳ only one unique solution.

So. $Ax=b$

↳ Only one unique solution exists.

Case 4: Not full rank, $r < m$ and $r < n$

Subcase 4.1: if b is in the column space of A .

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 2 & 6 \\ 3 & 6 & 3 & 9 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 12 \\ 18 \end{bmatrix}$$

Here

$$x_p = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \end{bmatrix}, x_n = c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -3 \\ 6 \\ 0 \\ 1 \end{bmatrix}$$

$$\hookrightarrow x_p = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad x_h = c_1 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\hookrightarrow$ represents infinitely many solutions depending on the value of c_1, c_2 and c_3 .

Subcase 4.2: if b is not in the column space of A ,

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 6 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$\hookrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 4 & 2 & 2 \\ 3 & 6 & 3 & 5 \end{array} \right]$$

$$\text{RREF} \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{array} \right] \rightarrow (0 \ 0 \ 0 \mid 2) \text{ indicates an inconsistency.}$$

This states that the system has no solution.

So, a matrix that is not full rank
will either have 0 or infinitely many solutions.

Summary:

i If $n = m$, $n < m \rightarrow 0$ or 1 solution

ii If $n = m$, $n < n \rightarrow \infty$ solutions

iii If $n = m = n \rightarrow 1$ solution

iv If $n < m$, $n < n \rightarrow 0$ or ∞ solutions.